

Proof term

Input

```
fun (A : R -> Prop) (x y : R) (H : x
is_a_max_of A) =>
match H with | conj x0 x1 =>
  (fun (HxA : A x) (Hx : up_bounds A x) (H0
: y is_a_max_of A) => match H0 with
| conj x2 x3 => (fun (HyA : A y) (Hy :
up_bounds A y) => Rle_antisym x y (Hy x HxA)
(Hx y HyA)) x2 x3 end) x0 x1 end
: forall (A : R -> Prop) (x y : R),
x is_a_max_of A -> y is_a_max_of A -> x = y

Lemma unique_max (A : R -> Prop) (x y : R) :
  x is_a_max_of A -> y is_a_max_of A -> x =
y.
```

Statement

Goals

```
F: CompleteOrderedField
A: R -> Prop
x, y: R
-----
x is_a_max_of A -> y is_a_max_of A -> x = y
```

Prediction

Think block 1

Okay, let's try to transform this proof term into a sequence of coq tactics. First let's write down the hypotheses, and the initial goal (after the "|-" symbol) given by the coq proof assistant: [...]. A good first step would be `intros [...]`. The goal will then become `y is\_a\_max\_of A -> x = y`.*)

Script block 1

```
intros [HxA Hx].
```

Think block 2

Alright, after the first step, my proof state should look like this: [...]. So, I'll use the `intros` again: `intros [HyA Hy]`. This will give me two new hypotheses, `HyA : A y` and `Hy : forall z : R, A z -> z <= R y`. The goal will be updated to `x = y`. [...]

Script block 2

```
intros [HyA Hy].
```